In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("retail_sales_dataset.csv")

print("Dataset imported successfully!")
print("Shape:", df.shape)

df.head()
```

```
Dataset imported successfully!
Shape: (1000, 9)
```

Out[1]:

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Pric
0	1	2023-11-24	CUST001	Male	34	Beauty	3	
1	2	2023-02-27	CUST002	Female	26	Clothing	2	5
2	3	2023-01-13	CUST003	Male	50	Electronics	1	
3	4	2023-05-21	CUST004	Male	37	Clothing	1	5
4	5	2023-05-06	CUST005	Male	30	Beauty	2	

In [2]:
```
print("Dataset Information:")
df.info()

print("\nColumn Names:")
print(df.columns.tolist())

print("\nStatistical Summary:")
display(df.describe())

print("\nMissing Values:")
print(df.isnull().sum())
```

```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Transaction ID    1000 non-null   int64
 1   Date              1000 non-null   object
 2   Customer ID       1000 non-null   object
 3   Gender            1000 non-null   object
 4   Age               1000 non-null   int64
 5   Product Category  1000 non-null   object
 6   Quantity          1000 non-null   int64
 7   Price per Unit    1000 non-null   int64
 8   Total Amount      1000 non-null   int64
dtypes: int64(5), object(4)
memory usage: 70.4+ KB

Column Names:
['Transaction ID', 'Date', 'Customer ID', 'Gender', 'Age', 'Product Category'
'Quantity', 'Price per Unit', 'Total Amount']

Statistical Summary:
```

	Transaction ID	Age	Quantity	
count	1000.000000	1000.000000	1000.000000	10
mean	500.500000	41.392000	2.514000	1
std	288.819436	13.681433	1.132734	1
min	1.000000	18.000000	1.000000	
25%	250.750000	29.000000	1.000000	
50%	500.500000	42.000000	3.000000	
75%	750.250000	53.000000	4.000000	3
max	1000.000000	64.000000	4.000000	5

```
Missing Values:
Transaction ID      0
Date                0
Customer ID         0
Gender              0
Age                 0
Product Category    0
Quantity            0
Price per Unit      0
Total Amount        0
dtype: int64
```

In [4]:
```python
print("Relevant Variables:")

print("Product Categories:")
print(df["Product Category"].unique())

print("\nTotal Amount:")
print(df["Total Amount"].head())
```

```
print("\nQuantity:")
print(df["Quantity"].head())
```

```
Relevant Variables:
Product Categories:
['Beauty' 'Clothing' 'Electronics']

Total Amount:
0     150
1    1000
2      30
3     500
4     100
Name: Total Amount, dtype: int64

Quantity:
0    3
1    2
2    1
3    1
4    2
Name: Quantity, dtype: int64
```

In [5]:
```
category_sales = df.groupby(
    "Product Category"
)["Total Amount"].sum()

print("Total Sales by Product Category:")
print(category_sales)
```

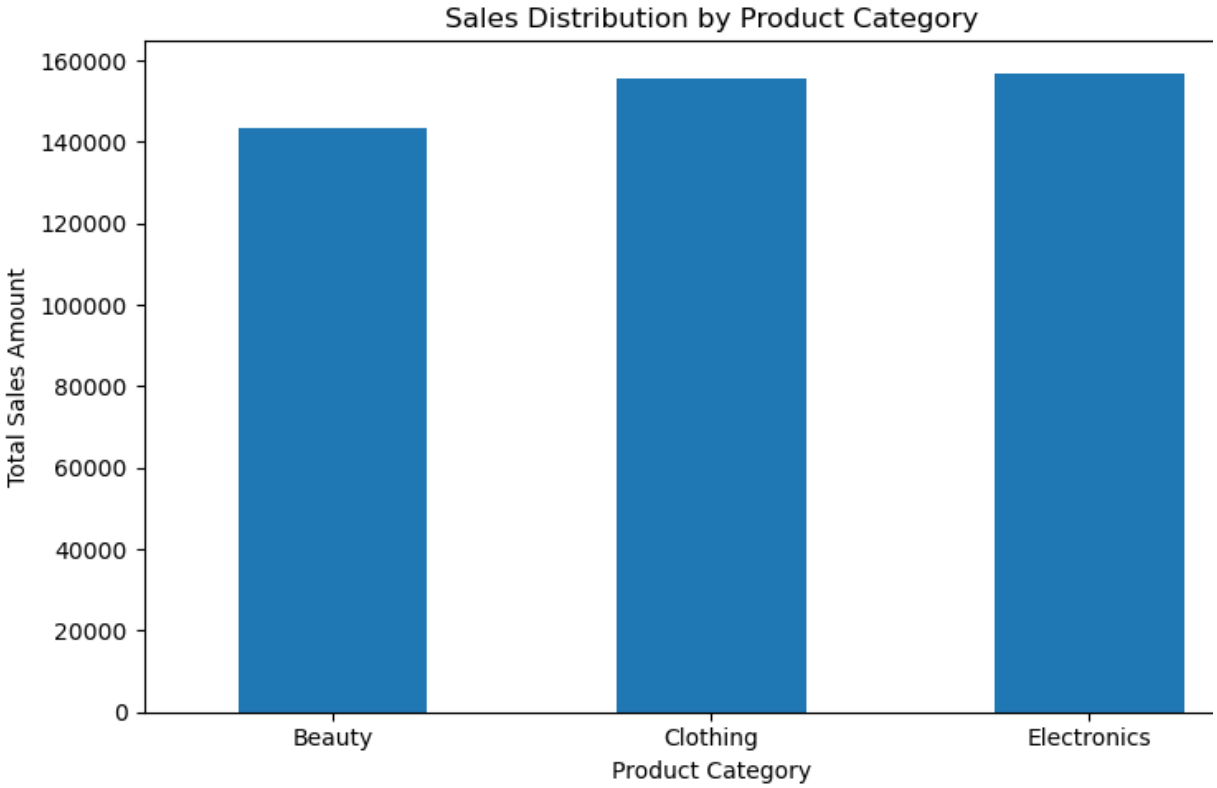
```
Total Sales by Product Category:
Product Category
Beauty         143515
Clothing       155580
Electronics    156905
Name: Total Amount, dtype: int64
```

In [6]:
```
plt.figure(figsize=(8, 5))

category_sales.plot(kind="bar")

plt.xlabel("Product Category")
plt.ylabel("Total Sales Amount")
plt.title("Sales Distribution by Product Category")

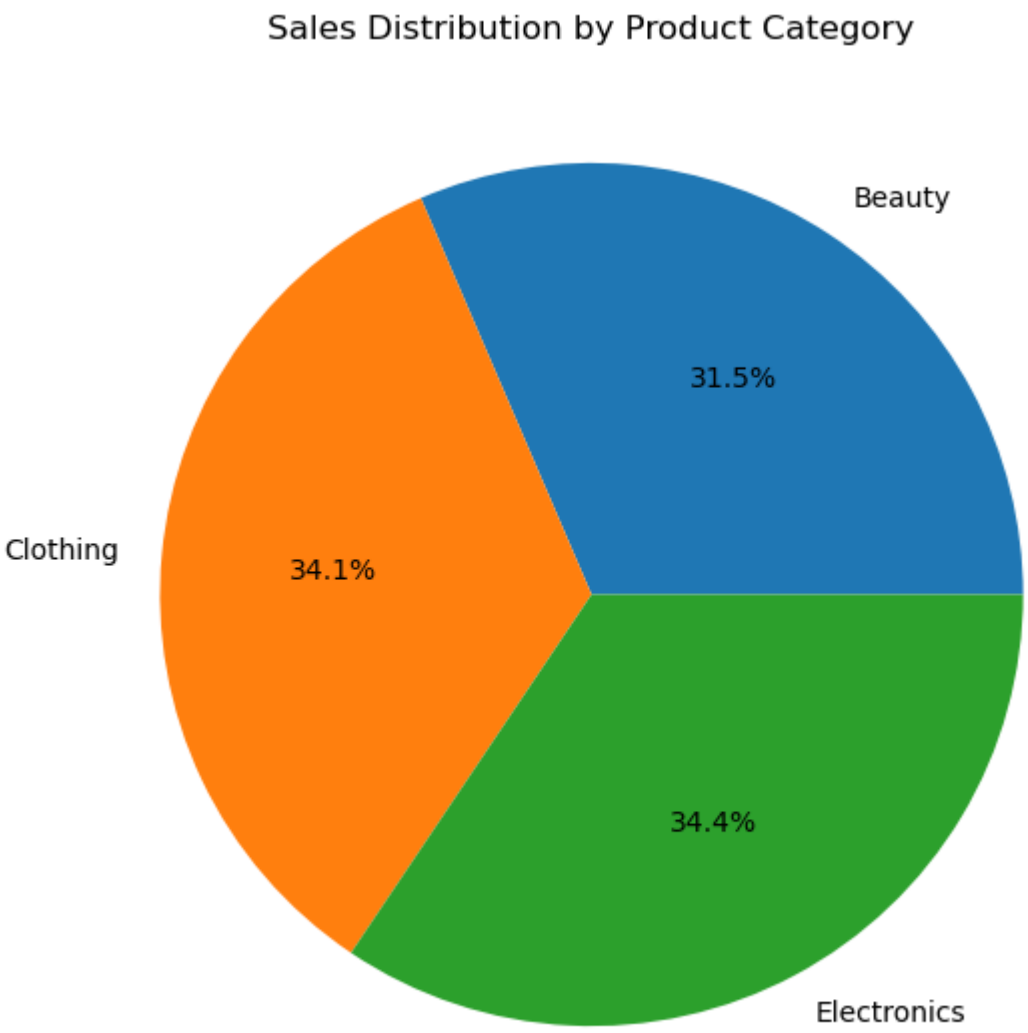
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



In [7]:
```
plt.figure(figsize=(7, 7))

plt.pie(
    category_sales,
    labels=category_sales.index,
    autopct="%1.1f%%"
)

plt.title("Sales Distribution by Product Category")
plt.show()
```



In [8]:
```
category_sales = category_sales.sort_values(
    ascending=False
)

print("Sales Ranking:")
print(category_sales)

print("\nTop-Performing Product Category:")
print(category_sales.index[0])

print("Highest Sales Amount:")
print(category_sales.iloc[0])
```

```
Sales Ranking:
Product Category
Electronics    156905
Clothing       155580
Beauty         143515
Name: Total Amount, dtype: int64

Top-Performing Product Category:
Electronics
Highest Sales Amount:
156905
```

In [9]:
```
gender_category_sales = df.groupby(
    ["Gender", "Product Category"]
)["Total Amount"].sum().unstack()

print("Sales by Gender and Product Category:")
display(gender_category_sales)
```

```
Sales by Gender and Product Category:
```

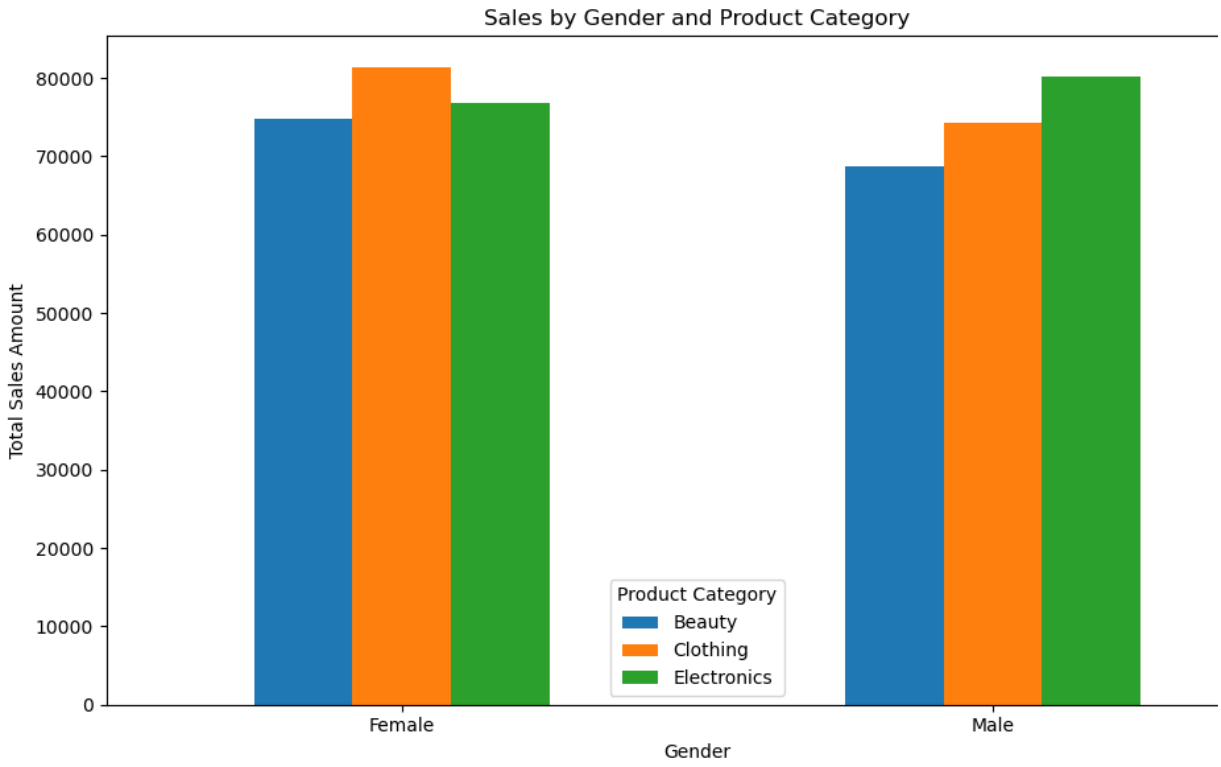
Product Category	Beauty	Clothing	Electronics
Gender			
Female	74830	81275	76735
Male	68685	74305	80170

In [10]:
```
gender_category_sales.plot(
    kind="bar",
    figsize=(10, 6)
)

plt.xlabel("Gender")
plt.ylabel("Total Sales Amount")
plt.title("Sales by Gender and Product Category")

plt.xticks(rotation=0)
plt.legend(title="Product Category")

plt.tight_layout()
plt.show()
```



In [11]:
```
print("========== SALES SUMMARY ==========")

print("\nTotal Sales:")
print(df["Total Amount"].sum())
```

```
print("\nSales by Product Category:")
print(category_sales)

print("\nTop Performing Category:")
print(category_sales.idxmax())

print("\nHighest Sales:")
print(category_sales.max())
```

```
========== SALES SUMMARY ==========

Total Sales:
456000

Sales by Product Category:
Product Category
Electronics    156905
Clothing       155580
Beauty         143515
Name: Total Amount, dtype: int64

Top Performing Category:
Electronics

Highest Sales:
156905
```

In [ ]: